5.1.4 Wrap-Up: Cool Down

1. Maggie and Naomi both constructed a triangle using the side lengths shown.

   - $CN = 2.5$ inches (in.)
   - $NK = 3.75$ in.
   - $CK = 5.25$ in.

   Explain whether it is possible to find out if the two triangles Maggie and Naomi created will be congruent.

Unit 5, Lesson 2: Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence

Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.5.1 Construct a copy of a segment or an angle.

Learning Targets

- ☐ I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.

5.2.1 Warm-Up: Geometry in Art

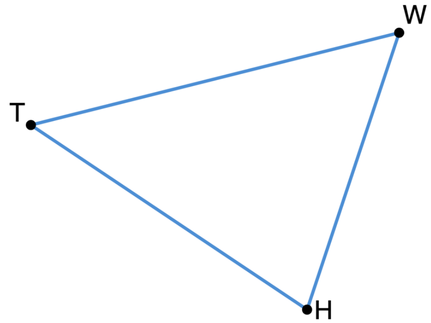
A Pysanka is a decorated egg that is traditional in Ukraine. The decoration on the eggs is created using a wax resist (batik) method. Families often have their own unique pattern. The largest Pysanka is in Vegreville, Canada. It weighs $5{,}000$ pounds, is 31 feet (ft) tall, and 18 ft wide. A traditional Pysanka egg is 2.5 inches tall and 1.75 inches wide.

1. What do you notice about Pysanka eggs?

2. Estimate the number of traditional Pysanka eggs that will fit into the largest Pysanka. Show your work or explain how you found your estimate.

5.2.2 Exploration: Congruence Relationships

1. Triangle TWH is shown.



   a. Construct ΔXMP so that $\overline{XM} \cong \overline{TW}$, $\overline{PM} \cong \overline{HW}$, and $\angle TWH \cong \angle PMX$ using the given steps.

Construction	Steps
	1. Copy $\overline{TW}$ using a compass to construct $\overline{XM}$ using the orange line so that $\overline{XM} \cong \overline{TW}$.
	2. Copy $\angle TWH$ to construct $\angle PMX$ so that $\angle TWH \cong \angle PMX$.
	3. Copy $\overline{HW}$ to construct $\overline{PM}$ so that $\overline{PM} \cong \overline{HW}$.
	4. Draw $\overline{PX}$.

   b. Use patty paper to determine if ΔXMP is congruent to ΔTWH.

   c. Discuss the following with your partner: If two sides of a triangle and their included angles are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

2. Construct ΔKSB so that $\overline{KS} \cong \overline{TH}$, $\overline{SB} \cong \overline{TW}$, and $\angle TWH \cong \angle SBK$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{KS}$ using the red line so that $\overline{TH} \cong \overline{KS}$.
	2. Copy $\overline{TW}$ to construct $\overline{SB}$ so that $\overline{TW} \cong \overline{SB}$.
	3. Copy $\angle TWH$ to construct $\angle SBK$ so that $\angle TWH \cong \angle SBK$.
	4. Draw $\overline{BK}$.

a. Use patty paper to determine if ΔKSB is congruent to ΔHTW.

b. Discuss the following with your partner: If two sides of a triangle and a non-included angle are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

3. Construct ΔRYV so that $\overline{VY} \cong \overline{TH}$, $\angle WTH \cong \angle RVY$, and $\angle WHT \cong \angle RYV$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{VY}$ using the green line so that $\overline{TH} \cong \overline{VY}$.
	2. Copy $\angle WHT$ to construct $\angle RYV$ so that $\angle WHT \cong \angle RYV$.
	3. Copy $\angle WTH$ to construct $\angle RVY$ so that $\angle WTH \cong \angle RVY$.
	4. Label the intersection of the sides of the two angles point R.

a. Use patty paper to determine if ΔRYV is congruent to ΔWHT.

b. Discuss the following with your partner: If two angles of a triangle and their included side are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

5.2.3 Bring It Together: Triangle Congruence

1. Complete the statements.

   a. ΔXMP was created using ○ 1 ○ 2 ○ 3 side(s) and ○ 1 ○ 2 ○ 3 angle(s).

   The construction showed that it is ○ possible ○ not possible to assume that if the ○ 1 ○ 2 ○ 3 side(s) of a triangle and their ○ adjacent angle(s) ○ included angle(s) ○ opposite angle(s) are congruent, then the triangles are congruent.

   b. ΔKSB was created using ○ 1 ○ 2 ○ 3 side(s) and ○ 1 ○ 2 ○ 3 angle(s). The construction showed that it is ○ possible ○ not possible to assume that if the ○ 1 ○ 2 ○ 3 side(s) of a triangle and the ○ included angle(s) ○ non-included angle(s) are congruent, then the triangles are congruent.

   c. ΔRYV was created using ○ 1 ○ 2 ○ 3 side(s) and ○ 1 ○ 2 ○ 3 angle(s).

   The construction showed that it is ○ possible ○ not possible to assume that if the ○ 1 ○ 2 ○ 3 side(s) of a triangle and its ○ adjacent angle(s) ○ included angle(s) ○ opposite angle(s) are congruent, then the triangles are congruent.

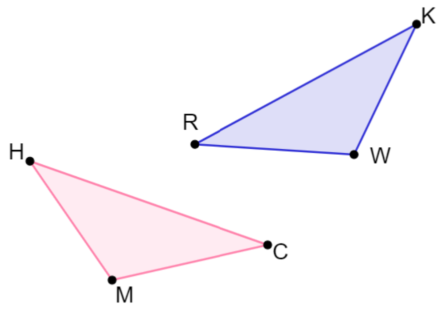
Side-Angle-Side (SAS) Congruence Postulate

If two sides and the included angle of one triangle are congruent to two sides and the included angle of a second triangle, respectively, then the two triangles are congruent.

Angle-Side-Angle (ASA) Congruence Theorem

If two angles and an included side of one triangle are congruent respectively to two angles and an included side of a second triangle, then the two triangles are congruent.

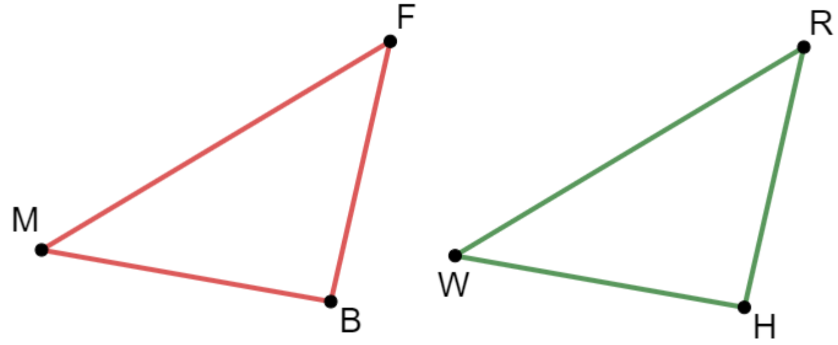
2. Triangles HCM and RKW are shown. Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate to show that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{MC} \cong \overline{RW}$, $\overline{HC} \cong \overline{KR}$, $\angle MCH \cong \angle WRK$		
$\overline{MC} \cong \overline{KW}$, $\overline{HM} \cong \overline{WR}$, $\overline{HC} \cong \overline{KR}$		
$\overline{KW} \cong \overline{HM}$, $\angle KRW \cong \angle CHM$, $\overline{MC} \cong \overline{RW}$		
$\overline{RK} \cong \overline{CH}$, $\angle MCH \cong \angle WRK$, $\angle MHC \cong \angle RKW$		
$\angle RKW \cong \angle CHM$, $\angle RWK \cong \angle CMH$, $\overline{KW} \cong \overline{HM}$		

5.2.4 Wrap-Up: Ticket Out the Door

Triangles BFM and WRH are shown.



1. Determine if the given information is sufficient to determine that the triangles are congruent. If so, state the congruence theorem or postulate to show that.

Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{FB} \cong \overline{HW}$, $\overline{FM} \cong \overline{RW}$, $\angle BFM \cong \angle HWR$		
$\overline{FB} \cong \overline{RH}$, $\angle FBM \cong \angle RHW$, $\overline{MB} \cong \overline{WH}$		
$\overline{FB} \cong \overline{HW}$, $\overline{BM} \cong \overline{HR}$, $\overline{FM} \cong \overline{RW}$		